

Predictive Branch Communication Theory

An advanced speculative framework for shared-state branch encoding, branch residuals, and deviation-based field communication

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Status: This is a speculative theoretical framework. It is not established physics. It deliberately marks the line between accepted results and the extra assumption required for the theory: that a local trigger in one system can controllably change the locally observable outcome distribution of a distant coupled system without an ordinary light-speed signal path.

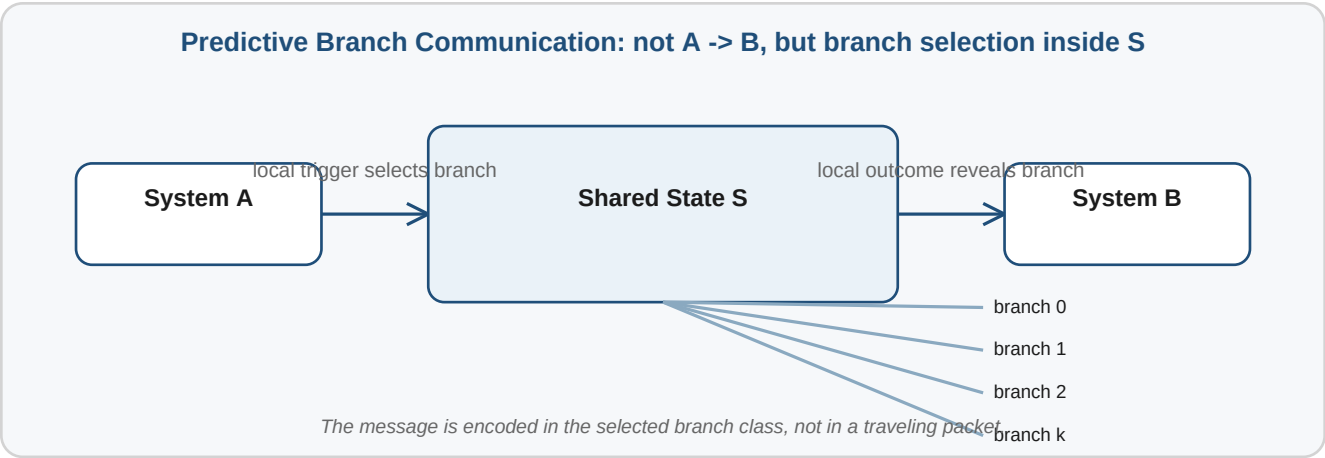


Figure 1. The core idea: a message is not transported as a packet. It is encoded as a selected branch class of a shared state.

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1. Executive summary

Predictive Branch Communication Theory, abbreviated PBC, is a speculative communication framework in which two distant systems do not exchange information by sending a conventional signal through intervening space. Instead, both systems are treated as local interfaces of a wider shared state, a state with a known correlation map, known layer hierarchy, and known rulebook for how uncertainty unfolds into measurable outcomes.

The message is not carried by a wave packet, photon, radio pulse, gravitational wave, or normal field disturbance. The message is encoded in the selection of one branch from a set of possible branches. System A applies a local trigger. If the theory is correct, that trigger selects or biases a branch class in the shared state. System B then observes its own local deviation from its expected path and decodes the branch class as information.

One-sentence theory: If two distant systems share a stable predictive field-correlation protocol, then a controlled local trigger in one system can encode information by selecting a branch of the shared uncertainty structure, while the other system decodes the message through a measurable branch residual in its own local evolution.

The framework becomes meaningful only if a hard operational criterion is met: A chosen trigger at A must change the standalone outcome distribution measured at B. In compact form, communication requires $P(Y_B | X_A = x1)$ to differ from $P(Y_B | X_A = x2)$, where X_A is the trigger chosen at A and Y_B is the locally observed outcome at B. If those distributions are identical, the apparent structure is correlation or prediction, not communication.

Communication threshold:

If $P(Y_B | X_A = x1) = P(Y_B | X_A = x2)$, capacity = 0.
If $P(Y_B | X_A = x1) \neq P(Y_B | X_A = x2)$, capacity may be positive.

The exact question becomes: how much information about X_A can B recover from Y_B ?

PBC is therefore best understood as a proposed channel model. It does not merely claim that distant systems are correlated. It claims that branch selection is controllable, locally readable, repeatable, and decodable.

2. Core thesis

The conventional communication picture is linear: a sender creates a signal, the signal propagates through a medium or field, and the receiver detects it after a time delay constrained by the propagation speed. PBC replaces that picture with a shared-state picture. In PBC, A and B are not isolated objects joined only by a traveling messenger. They are treated as partially exposed surfaces of a deeper joint state.

The central claim is not merely that A and B are correlated. It is that a local intervention at A can choose among possible future branches of the shared state, and B can infer that choice from its own local residual pattern.

Conventional signaling	Predictive Branch Communication
Information rides on a carrier through space.	Information is encoded by choosing a branch of a shared state-space.
Receiver waits for a signal to arrive.	Receiver detects a deviation from predicted local evolution.
Speed is tied to propagation through a medium or field.	Apparent speed comes from a pre-existing correlation protocol and branch selection.
The raw signal contains the message.	The branch residual, interpreted through the shared protocol, contains the message.

The phrase "predictive" matters because the receiver must possess a model of what should have happened in the absence of the selected branch. Without a prediction, there is no residual. Without a residual, there is no branch evidence. Without branch evidence, there is no decoding.

3. Why this is not ordinary signaling

Ordinary signaling has a transport structure. A cause is created at A, some physical carrier leaves A, traverses a path, and arrives at B. The resulting event at B is explained by propagation. PBC is instead a selection structure. A

trigger changes which branch of an already-existing joint protocol becomes realized. B does not receive the whole message as a payload. B reads the selected branch through its own local behavior.

Ordinary signal model:
A creates packet M -> packet travels -> B receives M

PBC model:
A selects branch k -> shared state S_k is realized -> B observes residual pattern $r_B(k)$ -> B decodes k

The receiver therefore needs a codebook. If branch k has no agreed interpretation, B may observe a strange event but cannot decode a message. A branch codebook is the mapping from branch classes to symbols.

This is the reason PBC can seem to carry more meaning than the raw trigger. The trigger is small, but it acts like a selector. The meaning is stored in the pre-existing map, not generated entirely by the trigger. A switch can be small and still choose a large outcome if the system already contains a structured set of alternatives.

4. The required extra assumption

The entire theory stands or falls on one additional assumption beyond accepted operational physics. The assumption is not "systems can be correlated." That is already familiar in many forms. The assumption is this:

PBC coupling assumption: For certain highly structured systems, a local trigger at A can controllably change the branch-dependent local outcome distribution at B without requiring an ordinary propagating signal from A to B.

This assumption is stronger than standard quantum entanglement. Standard entanglement can produce nonclassical correlations, but those correlations do not allow a user to choose a message at A and make B see a corresponding local distribution change by measurement alone. PBC explicitly asks what follows if such controllable branch coupling exists.

The minimal empirical requirement is not philosophical. It is statistical. For two triggers x_0 and x_1 at A, B must observe different distributions:

Required condition:
 $P(Y_B | X_A = x_0) \neq P(Y_B | X_A = x_1)$

where:
 X_A = controlled trigger chosen at A
 Y_B = local observation sequence recorded at B

This makes the framework testable in principle. It also makes it fragile: any result requiring later comparison with A before B can know anything is not PBC communication. It is only correlation analysis.

5. Ontology of the theory

A useful theory needs named objects. The following ontology separates the physical substrate from the communication role played by each part.

Object	Meaning in PBC	Operational question
System A	The sender-side local handle on the shared state.	Can A choose among triggers intentionally or randomly?
System B	The receiver-side local handle on the shared state.	Can B read branch-dependent local outcomes without receiving a conventional signal?
Shared state S	The deeper correlated state-space linking A and B.	Does S have stable, repeatable branch behavior?
Trigger X_A	A local perturbation, cue, orientation change, timing change, or field adjustment at A.	Does changing X_A alter B statistics?

Object	Meaning in PBC	Operational question
Branch K	A selected class of possible futures or dynamical basins.	Can K be distinguished from other branches?
Outcome Y_B	The observed local data at B.	Does Y_B carry information about X_A?
Prediction model M_B	B-side model of what should happen absent a branch selection.	Can B compute an expected path?
Branch residual r_B	Difference between B actual behavior and B expected behavior.	Can residual patterns be decoded as symbols?
Codebook C	Mapping from triggers/branches/residuals to symbols.	Can both sides use the same symbol map?

This ontology helps avoid a common mistake: treating every correlation as a message. A message requires agency over the input side and recoverability at the output side. Without both, the theory reduces to prediction, not communication.

6. Layer architecture

The theory uses a layered view because the proposed communication is not a single variable event. It depends on a substrate, a shared protocol, a trigger mechanism, branch dynamics, and observation.

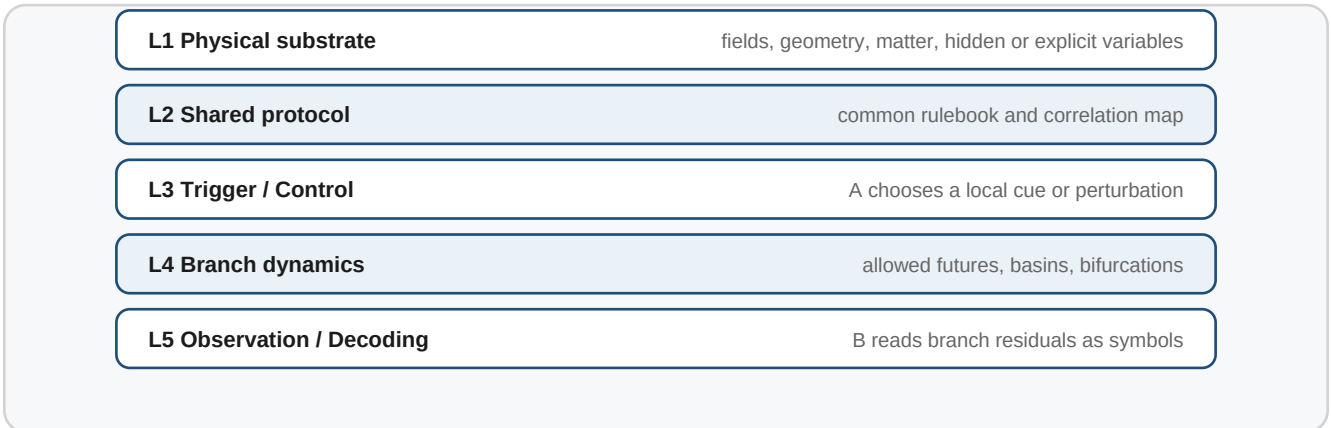


Figure 2. PBC layer hierarchy. A trigger becomes meaningful only when embedded in a shared protocol and read through an observation layer.

6.1 Substrate layer

The substrate is whatever physical or hypothetical structure carries the correlation: fields, geometry, matter distribution, black-hole spin/orientation parameters, dark-sector variables, hidden variables, holographic degrees of freedom, or another deeper state description. PBC does not require choosing one substrate at the start, but it requires that the substrate support stable correlations.

6.2 Shared protocol layer

The protocol layer is the rulebook. It defines which states are allowed, which transitions are likely, which branches exist, and how outcomes at B relate to triggers or branch classes at A. This protocol may be learned from long historical observation or derived from theory.

6.3 Trigger layer

A trigger is a local cue that changes the branch distribution. It could be an orientation adjustment, phase shift, timing injection, energy pulse, field perturbation, spin-like choice, or geometrical realignment. The theory requires the trigger to be selectable, not merely observed after the fact.

6.4 Branch dynamics layer

Branch dynamics describe how the shared state moves from uncertainty into a particular branch class. This includes thresholds, bifurcations, attractors, memory effects, resonance, amplification, and hysteresis. A branch may be a

single outcome or a family of related outcomes.

6.5 Observation and decoding layer

B observes local outcomes and compares them against the expected path. The difference is the branch residual. Decoding converts the residual into a symbol, a bit, or a higher-level message.

7. Formal model

A minimal PBC model can be written as a controlled stochastic channel with a shared hidden or explicit state. The notation below is deliberately simple and operational.

Variables:

S_t = shared state at time t
 $X_A(t)$ = trigger chosen locally at A
 K_t = selected branch class
 $Y_B(t)$ = local observation at B
 M_B = B-side predictive model
 $r_B(t)$ = branch residual at B

Branch selection:

$K_t \sim P(K | S_t, X_A(t))$

B observation:

$Y_B(t) \sim P(Y_B | K_t, S_t, \text{local_B_conditions})$

Residual:

$r_B(t) = Y_B(t) - E_{M_B}[Y_B(t) | \text{no selected trigger}]$

The communication object is the conditional distribution $P(Y_B | X_A)$. If different inputs at A generate distinguishable distributions at B, the channel can carry information. If all inputs generate the same B-side distribution, then the channel has zero communication capacity even if A and B are deeply correlated.

The communication object is the conditional channel $P(Y_B | X_A)$

	0.82	0.10	0.05	0.03
	0.10	0.75	0.10	0.05
	0.08	0.12	0.70	0.10

Rows are triggers selected at A; columns are branch-dependent outcomes observed at B.

Figure 3. Channel view of PBC. The matrix is not a data table from nature; it illustrates the required conditional structure.

7.1 Branch distinguishability

A branch is usable only if B can distinguish it from other branches under noise. The distance between two B-side distributions can be measured by total variation distance, Kullback-Leibler divergence, mutual information, classification accuracy, or another appropriate statistic.

For two triggers x_0 and x_1 :

$D = \text{distance}(P(Y_B | x_0), P(Y_B | x_1))$

If $D = 0$: no branch signal.

If $D > 0$ but small: repeated trials and error correction are required.

If D is large: direct decoding may be possible.

7.2 Channel capacity

The Shannon-style version of the question is: how much information about A's chosen trigger can B recover from its local observations? The answer depends on the input alphabet, branch noise, output overlap, and decoder quality.

PBC capacity question:

$C_{PBC} = \max \text{ over input choices } P(X_A) \text{ of } I(X_A; Y_B)$

where $I(X_A; Y_B)$ is the mutual information between chosen triggers at A and local observations at B.

This is not proof that such a channel exists. It is the correct way to measure it if it does.

A million possible branches do not create a million-symbol channel unless A can select among them and B can distinguish them. The usable alphabet is the set of controlled, distinguishable branch classes.

8. Branch residuals and deviation encoding

The branch residual is one of the most important concepts in the refined theory. B is not necessarily reading A directly. B is reading its own prediction error. If the prediction error has a stable relationship to A's trigger, then it functions as the received symbol.

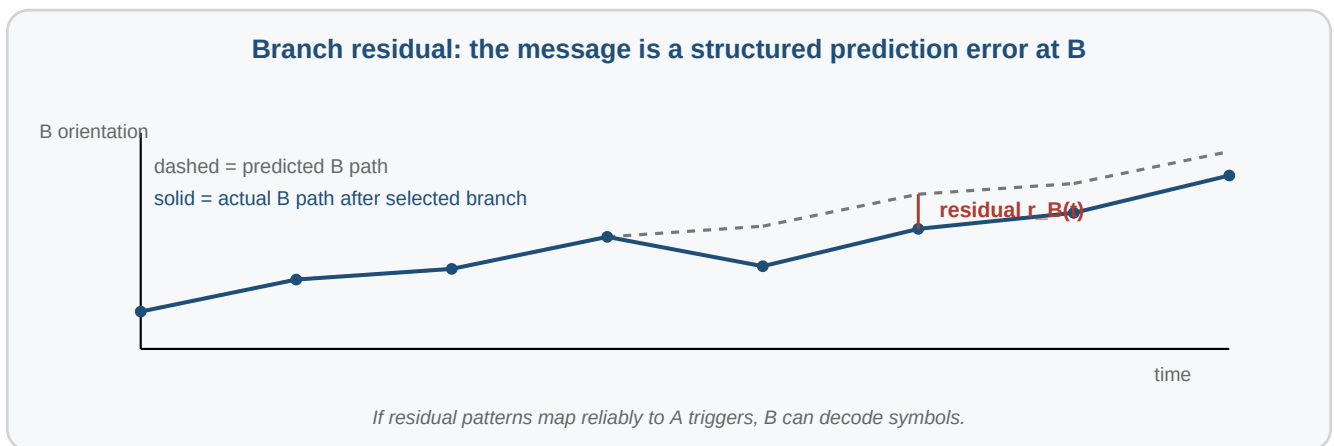


Figure 4. The branch residual is the gap between expected and observed B evolution.

8.1 Definition

Branch residual:

$$r_B(t) = \text{observed}_B(t) - \text{predicted}_B(t)$$

Symbolic branch residual:

R_B = pattern class of r_B over a time window

Decode rule:

message = decoder(R_B)

8.2 Why residuals matter

In many complex systems, raw readings are not informative because the system is already moving. The informative part is the deviation from expectation. If B was expected to rotate through 20 degrees but rotates through 16 degrees exactly when A was forced into a particular branch, the 4-degree residual may carry the symbol.

8.3 Residual codebook

A residual codebook maps branch residuals to symbols. For example, a negative amplitude residual may mean 0, a positive timing residual may mean 1, a phase residual may mean 2, and a frequency residual may mean 3. A richer codebook can combine amplitude, phase, timing, duration, and cross-field signatures.

A-side selected branch	Expected B residual pattern	Decoded symbol
K0: no perturbation	Residual near zero within noise limits	0 or idle
K1: extended orientation branch	Amplitude suppression at B	1
K2: compressed orientation branch	Amplitude expansion at B	2
K3: timing-offset branch	Phase or timing residual at B	3
K4: multi-field branch	Coupled amplitude and timing residual	control symbol

9. Branch alphabet and message design

For communication, the branch space must become an alphabet. This means triggers at A, branch classes in the shared state, residual classes at B, and symbols in the message must be mapped consistently.

Trigger -> Branch -> Residual -> Symbol

x0 -> K0 -> R0 -> bit 0
x1 -> K1 -> R1 -> bit 1
x2 -> K2 -> R2 -> symbol A
x3 -> K3 -> R3 -> symbol B

9.1 Alphabet requirements

A usable alphabet must satisfy four constraints. First, the sender must be able to choose each symbol-producing trigger. Second, each trigger must lead to a stable branch distribution. Third, each branch must produce a distinguishable residual at B. Fourth, the decoder must recover symbols above chance without later receiving the trigger record through ordinary communication.

9.2 Error correction

If branches overlap under noise, PBC would require repetition and coding, just like ordinary communication. A bit could be sent by repeating a branch selection many times and using majority decoding. Higher-order messages could be encoded with parity checks, block codes, or timing windows.

This turns the theory from a dramatic one-shot effect into a practical statistical channel. The channel may be weak, slow, and noisy even if the underlying coupling is unusual.

9.3 Meaning amplification

The raw trigger may be small, but its informational effect can be large if the system has preloaded structure. This is not magic. It is similar in logic to a key selecting a room, a seed selecting a growth path, or a bit flipping a program branch. The trigger does not contain the full content; it selects from a pre-existing structured space.

10. Black-hole orientation example

The motivating example can be formalized using two extremely distant gravitational systems, labeled A and B. Treat A as a black-hole-like object with an orientation parameter, spin-axis parameter, jet-axis parameter, or another angular observable. Treat B as a distant system with its own angular, field, or dynamical response observable.

Clarification: The example uses angular degrees, not degrees Celsius. Celsius is a temperature unit. In this document, the relevant variable is orientation in degrees.

10.1 Normal historical relation

Suppose decades or centuries of observation establish a predictive relation. Under normal conditions, A evolves within one orientation range and B evolves within a corresponding response range.

Normal relation:

A orientation range: 0 to 50 degrees

B response range: 0 to 20 degrees

Learned predictive map:

$B_expected(t) = f(A_history, B_history, field_layers, geometry, phase, memory)$

10.2 Branch selection event

Now suppose a controlled local perturbation or naturally selectable trigger pushes A into an unusual branch. Instead of the expected 0-to-50-degree family, A enters a 0-to-70-degree family. Under PBC, this is not merely a local anomaly. It changes the selected branch class of the shared state.

Selected branch at A:

Expected A family: 0 to 50 degrees

Selected A family: 0 to 70 degrees

Predicted B under old branch: 0 to 20 degrees

Observed B under selected branch: 0 to 16 degrees

Branch residual at B: -4 degrees relative to expected range endpoint

10.3 What B actually reads

B does not necessarily know "A moved to 70 degrees" as raw content. B sees that its own state deviated from its expected path. If the shared codebook says that a negative amplitude residual of this kind maps to branch K1, then B decodes K1.

The received information is not:

A rotated to 70 degrees.

The received information is:

B produced residual pattern R1, and R1 maps to branch K1 in the shared protocol.

10.4 The deeper claim

The deeper claim is that the two systems are polarized, phase-locked, or otherwise jointly constrained by a shared field-correlation structure. The trigger at A does not push a signal across the distance. It changes the branch selected by the wider structure, and B's local evolution reflects that branch.

This is the strongest version of the user-originated idea: information is encoded as controlled deviation from a predictive relation, not as a packet crossing light-years.

11. Recursive field-correlation interpretation

The phrase "recursive loop" can be made precise. A recursive predictive system is one whose current state is partly shaped by its past states, its expected future paths, and feedback among layers. In PBC, A and B share not merely a single correlation but a hierarchy of such loops.

11.1 Correlation map

A correlation map is a learned or theoretical model relating variables across A and B. In the advanced version, the map is not a simple linear equation. It is a multi-layer object with memory, thresholds, attractors, and cross-field dependencies.

Correlation map components:

1. Slow variables: mass, angular momentum, geometry, long-term field configuration.
2. Medium variables: phase, orientation, periodicity, orbital cycles, accretion patterns.
3. Fast variables: spikes, bursts, microstructure, local perturbations.
4. Hidden or latent variables: unobserved but inferable states.
5. Protocol variables: branch rules and allowed transition paths.

11.2 Recursive memory

Recursive memory means the response depends on history. A branch may not be selected by one instantaneous value alone. It may depend on the sequence of prior states. This creates richer coding possibilities because the message can be embedded in timing, pattern, and path-dependence.

11.3 Chain-reaction amplification

A tiny cue can become measurable if it enters a sensitive region of the state-space. Near a threshold, bifurcation, resonance, or unstable equilibrium, a small change can select a different macroscopic path. In PBC, the trigger works by choosing among already-available paths, not by creating a full macroscopic event from nothing.

11.4 Prediction-error communication

At the receiver side, communication appears as structured surprise. The system predicts one future, observes another, and the difference is not random. The residual has a code. That code is the message.

12. Relation to standard physics

PBC must be separated carefully from established physics. Several accepted theories contain ideas that sound similar but do not grant controllable faster-than-light communication.

12.1 Bell correlations and entanglement

Bell-theorem experiments show that quantum correlations cannot be reproduced by theories satisfying certain local realist conditions. The 2022 Nobel Prize in Physics recognized experiments with entangled photons, violations of Bell inequalities, and foundations of quantum information. This supports nonclassical correlation, not a user-controlled superluminal message channel.

12.2 No-signalling boundary

The standard no-signalling boundary is the operational rule that choices at A cannot be used to change the standalone statistics observed at B. Quantum field theory and ordinary quantum mechanics preserve this boundary in their standard form. PBC is explicitly a speculative framework about what would follow if that boundary were broken by a controllable shared branch mechanism.

12.3 Gravity and propagation speed

Under general relativity, gravitational changes are not treated as instantaneous signals. Gravitational waves are propagating spacetime disturbances, and modern gravitational-wave observations support propagation at essentially the speed of light. Therefore, if PBC uses gravitational language, it must either be a thought experiment, a new nonstandard coupling, or a theory about pre-existing global constraints rather than ordinary gravitational-wave propagation.

12.4 Why the distinction matters

If a proposed PBC effect is later explained by normal light-speed gravitational waves, electromagnetic leakage, shared environmental noise, data-analysis bias, or ordinary delayed comparison, then it is not PBC. PBC requires local readability at B from B-side data alone.

Phenomenon	Accepted status	Does it alone imply PBC?
Ordinary correlation	Common in many systems	No. Correlation is not controlled communication.

Phenomenon	Accepted status	Does it alone imply PBC?
Quantum entanglement	Established	No. Standard entanglement obeys no-signalling.
Bell inequality violation	Experimentally supported	No. It rules out certain local realist models but does not provide a message channel.
Gravitational waves	Observed by LIGO/Virgo/KAGRA-era astronomy	No. They propagate rather than act as instantaneous controllable branch channels.
PBC controllable branch coupling	Speculative	Yes, if $P(Y_B X_A)$ changes and B can decode it locally.

13. Causality problems and possible guardrails

A controllable instantaneous channel creates serious causality issues. If observers in different inertial frames disagree about time order, then a two-way faster-than-light channel can generate paradoxes under standard relativity. PBC therefore needs a causality guardrail if it is to be internally consistent.

13.1 The paradox risk

If A can send to B instantaneously and B can send back under the same rule, then some frames may describe the return message as arriving before the original trigger. This can create causal loops. Any serious PBC-like theory must explain why such loops do not occur or why they are globally consistent.

13.2 Possible guardrails

Guardrail	Idea	Cost
Preferred frame	Branch updates are instantaneous only in one cosmic reference frame.	Breaks or modifies relativistic symmetry.
One-way accessibility	Only certain systems can read branch residuals; reciprocal signalling is blocked.	Limits communication usefulness.
Probabilistic censorship	Attempts to create paradoxes cause noise to rise and capacity to fall to zero.	Requires a new rule.
Global consistency	Only self-consistent branch histories can be realized.	Turns the theory into a constraint-satisfaction model.
No controllable macroscopic choice	Coupling exists but cannot be used to freely select messages.	Preserves causality but destroys communication.

The last option is especially important: nature may allow deep correlations but block controllable selection. This is essentially the difference between impressive correlation and usable communication.

14. Experimental proof ladder

A theory like PBC must be built through increasingly strict tests. The proof ladder below starts with weak evidence and ends with true communication evidence.

Level	Test	Why it matters
1. Correlation	Show A and B have stable historical correlation.	Establishes a shared predictive relation.
2. Prediction	Build a model that predicts B from historical variables.	Creates a baseline for residual detection.
3. Trigger choice	Randomly choose triggers at A under blinded conditions.	Prevents after-the-fact pattern selection.
4. Distribution shift	Show B-side outcomes change with A triggers.	Tests $P(Y_B X_A)$ difference.
5. Local decoding	B predicts A trigger from B data alone.	Creates communication evidence.

Level	Test	Why it matters
6. Exclusion	Rule out light-speed signals, leakage, timing bias, and shared noise.	Protects against ordinary explanations.
7. Capacity estimate	Estimate bits per trial or bits per unit time.	Converts anomaly into a channel model.

14.1 Minimal blind protocol

- Blind test:
1. A randomly chooses trigger 0 or 1 using a pre-registered random schedule.
 2. B has no access to A schedule through ordinary channels.
 3. B records local outcomes during fixed windows.
 4. A decoder predicts the trigger from B data alone.
 5. Success is measured against chance, with corrections for multiple testing.
 6. The experiment is repeated with shielding, distance changes, timing randomization, and independent teams.

14.2 Astronomical version

An astronomical version would be much harder because humans cannot easily control a black hole. But one could search for natural trigger-like events: flares, jet-axis changes, accretion transitions, quasi-periodic oscillation shifts, or merger-related state changes. The difficulty is that natural events do not prove communication unless the input can be selected or predicted independently.

14.3 Laboratory analog version

A practical first step would not use black holes. It would use controllable coupled systems: resonators, nonlinear oscillators, plasma systems, quantum-optical systems, superconducting circuits, or field-sensitive chaotic systems. These would not prove cosmic PBC, but they could test the mathematical structure of branch residual encoding.

15. Failure modes and null hypotheses

PBC is vulnerable to many false positives. A rigorous framework must name them directly.

Failure mode	Description	Defense
Ordinary propagation	A normal signal traveled from A to B.	Use timing, shielding, distance, and known propagation models.
Shared environmental noise	Both systems responded to a third factor.	Monitor environment and use randomized controls.
Data mining	Patterns were selected after seeing the data.	Pre-register hypotheses and analysis code.
Clock bias	Timing alignment created artificial correlation.	Use independent clocks and blinded timestamps.
Model overfitting	Prediction model learned noise.	Holdout tests and independent replication.
Delayed comparison	B only learns after comparing with A later.	Require local decoding before comparison.
Natural correlation	A and B are correlated but A has no control.	Require selected triggers.

The strongest falsifier is simple: if A triggers do not change B-side standalone distributions, the communication channel does not exist. The theory may remain an interesting metaphor for prediction, but not a physical communication model.

16. Predictions of the framework

If PBC is real in any physical system, several signatures should appear. These are not guaranteed by existing physics; they are proposed signatures of the speculative mechanism.

- **Trigger-dependent residuals:** Different A-side triggers produce different B-side residual distributions.
- **Protocol specificity:** The effect is strongest in systems with matched structure, shared history, resonance, or phase-locking.
- **Threshold behavior:** Effects appear near branch boundaries, bifurcations, or critical points rather than across all states.
- **Noise-sensitive capacity:** The channel works statistically and may require repeated trials or error correction.
- **Layer coupling:** A perturbation at one layer produces a predictable cascade across other layers at B.
- **Residual code stability:** The same trigger class produces the same residual class under comparable conditions.
- **Control collapse under paradox attempts:** If causality guardrails exist, deliberate paradox-like protocols may destroy channel reliability.

A failed prediction also matters. If residuals cannot be mapped to triggers better than chance under blinded conditions, PBC communication is not present in that system.

17. Research roadmap

The theory should be developed in stages. Jumping directly to cosmic claims is weak. The advanced route is to build the channel mathematics, test analog systems, and only then discuss astrophysical interpretations.

Phase	Goal	Deliverable
Phase 1: Formalization	Define variables, branch alphabet, residuals, and channel metrics.	Mathematical white paper.
Phase 2: Simulation	Create artificial shared-state models where branch selection is known.	Simulated capacity curves and decoder tests.
Phase 3: Analog experiment	Test residual decoding in controlled nonlinear systems.	Blind experimental dataset.
Phase 4: Physical exclusion	Rule out ordinary leakage and shared noise.	Independent replication under controls.
Phase 5: Astrophysical search	Look for branch-residual signatures in long-term field data.	Pre-registered observational analysis.
Phase 6: Theory integration	Develop or reject candidate substrates.	Causality-consistent model or formal no-go result.

17.1 Simulation model

A toy model can be created with a shared latent state S , controlled triggers X_A , branch classes K , and observed residuals Y_B . By tuning noise, branch overlap, and trigger strength, one can study when decoding becomes possible. This does not prove nature works this way; it clarifies the mathematical requirements.

Toy simulation:

S_t evolves recursively.
A chooses X_A in $\{0,1\}$.
 K_t is selected by $P(K | S_t, X_A)$.
B observes $Y_B = g(K_t, S_t) + \text{noise}$.
Decoder estimates X_A from Y_B .
Capacity is estimated from decoding accuracy or mutual information.

17.2 Candidate analog systems

Good analog systems should have strong memory, nonlinear thresholds, controllable perturbations, measurable residuals, and a known model. Examples include coupled oscillators, nonlinear optical cavities, plasma modes, chaotic electronic circuits, and engineered networks with shared predictive models. These analogs test the communication architecture even if they do not violate light-speed causality.

18. Refined final formulation

The most precise version of the theory is the following:

Predictive Branch Communication Theory: Two distant systems can act as local interfaces of a shared branching state. If a selectable local trigger at System A changes the conditional distribution of branch-dependent local outcomes at System B, and if System B can decode that distribution through a pre-existing predictive protocol without ordinary signal comparison, then information has been communicated through branch selection rather than through a traveling carrier.

An even shorter technical version is:

PBC exists for a pair (A,B) only if there is at least one pair of A-side triggers x_i and x_j such that:

$$P(Y_B | X_A = x_i) \neq P(Y_B | X_A = x_j)$$

and B can infer X_A from Y_B above chance before any ordinary information about X_A arrives.

This formulation keeps the theory honest. It neither dismisses the idea too early nor allows it to hide inside vague words like connection, resonance, energy, or uncertainty. It states exactly what must be observed.

19. Glossary

Term	Definition
Branch	A class of possible future states or dynamical paths selected from a shared state-space.
Branch alphabet	A set of controlled, distinguishable branches used as communication symbols.
Branch residual	The difference between B actual behavior and B predicted behavior under the baseline model.
Branch selection	The process by which a trigger biases or chooses one branch class.
Codebook	A mapping from trigger classes and residual patterns to symbols.
Conditional distribution	A probability distribution of B outcomes given a particular A trigger.
Controllability	The ability of A to choose the branch-producing trigger rather than merely observe it.
Local decoding	B infers the message from B-side data alone, without later comparison to A.
Prediction error	Observed behavior minus expected behavior. In PBC this becomes meaningful if structured by branch selection.
Shared protocol	The pre-existing rulebook that relates A triggers, branch classes, and B residuals.
State-space	The set of possible states and paths available to the coupled system.

20. References and grounding notes

These sources are not evidence for PBC as an established phenomenon. They provide the standard physics and information-theory boundaries that PBC would have to confront.

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Closing note

The most advanced form of the idea is not "gravity is instantly sending messages." That wording is too easy to attack and too weak scientifically. The advanced form is sharper: if a shared predictive state has controllable branch selection and locally readable residuals, then information can be encoded in branch realization. The central experiment is to test whether B-side distributions depend on A-side choices. Everything else is architecture around that one fact.

Core sentence to preserve: Information is not transported; it is selected. The receiver does not receive a packet; it detects the branch residual of a shared predictive state.